* Non-Exact Differential Equations Reducible To Exact Form:-

- Sometimes a differential eq^{n} is not exact but can be made exact by multiplying with a suitable function. This function is known as Integrating Factor (I.F.) There may exists more than one integrating factor to a differential eq^{n}.

- We will discuss five different methods to find an I.F. to a non-exact differential eq^{n}.

$M\,dx + N\,dy = 0$

1) Case I:-

If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$ (function of x alone) then

$I.F. = e^{\int f(x)\,dx}$.

- After multiplication with the I.F., the equation becomes exact and can be solved using the method of exact differential eq^{ns}.

Examples:-

① Solve $(x^2+y^2+3)\,dx - 2xy\,dy = 0$

→

$M = x^2+y^2+3, \quad N = -2xy.$

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$\frac{\partial M}{\partial y} = 2y, \quad \frac{\partial N}{\partial x} = -2y$

Here, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$.

∴ The eqⁿ is not exact.

Now, $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y + 2y}{-2xy} = \frac{4y}{-2xy} = \frac{-2}{x}$

- I.F. $= e^{\int \frac{-2}{x} dx} = e^{-2\log x} = e^{\log x^{-2}} = x^{-2} = \frac{1}{x^2}$.

- Multiplying the D.E. by the I.F.,

$\frac{1}{x^2}(x^2+y^2+3)\,dx - \frac{1}{x^2} 2xy\, dy = 0$

∴ $\left(1 + \frac{y^2+3}{x^2}\right) dx - \frac{2y}{x} dy = 0$

∴ $M_1 = 1 + \frac{y^2+3}{x^2}, \quad N_1 = -\frac{2y}{x}$

∴ $\frac{\partial M_1}{\partial y} = \frac{2y}{x^2}, \quad \frac{\partial N_1}{\partial x} = \frac{2y}{x^2}$.

Here, $\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$, the eqⁿ is exact.

∴ The general solⁿ is

$\int M_1\, dx + \int$ (Terms of N_1 not containing x) $dy = c$.
(y constant)

∴ $\int \left(1 + \frac{y^2+3}{x^2}\right) dx + \int 0\, dy = C$

∴ $x - \frac{y^2+3}{x} = C$

∴ $x^2 - y^2 - 3 = cx$.

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② $(xy^2 - e^{yx^3})\,dx - x^2y\,dy = 0$.

→

$M = xy^2 - e^{yx^3}$, $N = -x^2y$

$\frac{\partial M}{\partial y} = 2xy$, $\frac{\partial N}{\partial x} = -2xy$.

Here, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, the eq^n is not exact.

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2xy - (-2xy)}{-x^2y} = \frac{4xy}{-x^2y} = \frac{-4}{x}.$$

$$\text{I.F.} = e^{\int \frac{-4}{x}\,dx} = e^{-4\log x} = e^{\log x^{-4}} = \frac{1}{x^4}.$$

Multiplying the D.E. by the I.F.,

$$\frac{1}{x^4}(xy^2 - e^{yx^3})\,dx - \frac{1}{x^4}(x^2y)\,dy = 0$$

$$\therefore \left(\frac{y^2}{x^3} - \frac{e^{yx^3}}{x^4}\right)dx - \frac{y}{x^2}\,dy = 0$$

$M_1 = \frac{y^2}{x^3} - \frac{e^{yx^3}}{x^4}$, $N_1 = -\frac{y}{x^2}$

$\frac{\partial M_1}{\partial y} = \frac{2y}{x^3}$, $\frac{\partial N_1}{\partial x} = \frac{2y}{x^3}$

$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$.

∴ The eq^n is exact.

∴ The general solution is

$\int M_1\,dx + \int$ (Terms of N_1 not containing x) $dy = c$.

y-constant

$$\therefore \int \left(\frac{y^2}{x^3} - \frac{e^{yx^3}}{x^4}\right)dx + \int 0\,dy = c.$$

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$$\therefore -\frac{y^2}{2x^2} + \frac{1}{3}\int e^{y/x^3}\left(-\frac{3}{x^4}\right) dx = c$$

$$\therefore -\frac{y}{2x^2} + \frac{1}{3} e^{y/x^3} = c \qquad \left[\because \int e^{f(x)} f'(x)dx = e^{f(x)} + c\right]$$

② Case - II :-

If $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = f(y)$ (function of y alone) then

I.F. $= e^{\int f(y)\, dy}$

- After multiplying with the I.F., the eqⁿ becomes exact and can be solved using the method of exact differential equation.

* Examples :-

① Solve $(y^4+2y)\,dx + (xy^3+2y^4-4x)\,dy = 0$.

→

Here, $M = y^4+2y$, $N = xy^3+2y^4-4x$

$\frac{\partial M}{\partial y} = 4y^3+2$, $\frac{\partial N}{\partial x} = y^3 - 4$

Here, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

∴ The eqⁿ is not exact.

Now, $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{y^3-4-4y^3-2}{y^4+2y} = \frac{-6-3y^3}{y^4+2y} = \frac{-3(2+y^3)}{y(2+y^3)}$

$= -\frac{3}{y}$

$\therefore$ I.F. $= e^{\int -\frac{3}{y} dy} = e^{-3\log y} = e^{\log y^{-3}} = \frac{1}{y^3}$

- Multiplying the D.E. by the I.F.,

$\frac{1}{y^3}(y^4+2y)\,dx + \frac{1}{y^3}(xy^3+2y^4-4x)\,dy = 0$

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$\therefore M_1 = y + \frac{2}{y^2}, \quad N_1 = x + 2y - \frac{4x}{y^3}$

$\therefore \frac{\partial M_1}{\partial y} = 1 - \frac{4}{y^3}, \quad \frac{\partial N_1}{\partial x} = 1 - \frac{4}{y^3}$

Here, $\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$, the eq^n is exact.

$\therefore$ The general sol^n is

$\int_{y\text{-constant}} M_1\,dx + \int (\text{Terms of } N_1 \text{ not containing } x)\,dy = c$

$\therefore \int \left(y + \frac{2}{y^2}\right) dx + \int 2y\,dy = c$

$\therefore \left(y + \frac{2}{y^2}\right) x + y^2 = c$

(2) Solve $(2xy^4e^y + 2xy^3 + y)\,dx + (x^2y^4e^y - x^2y^2 - 3x)\,dy = 0.$

→

Here, $M = 2xy^4e^y + 2xy^3 + y, \quad N = x^2y^4e^y - x^2y^2 - 3x.$

$\frac{\partial M}{\partial y} = 8xy^3e^y + 2xy^4e^y + 6xy^2 + 1, \quad \frac{\partial N}{\partial x} = 2xy^4e^y - 2xy^2 - 3$

$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}.$

$\therefore$ The eq^n is not exact.

Here, $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{2xy^4e^y - 2xy^2 - 3 - 8xy^3e^y - 2xy^4e^y - 6xy^2 - 1}{2xy^4e^y + 2xy^3 + y}$

$= \frac{-4 - 8xy^2 - 6xy^3e^y}{2xy^4e^y + 2xy^3 + y}$

$= \frac{-4(1 + 2xy^2 + 2xy^3e^y)}{y(1 + 2xy^2 + 2xy^3e^y)}.$

$= -\frac{4}{y}.$

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$$I.F. = e^{\int -\frac{4}{y}\,dy} = e^{-4\log y} = e^{\log y^{-4}} = \frac{1}{y^4}.$$

- Multiplying the DE by the I.F.,

$$\frac{1}{y^4}(2xy^4e^y + 2xy^3 + y)\,dx + \frac{1}{y^4}(x^2y^4e^y - x^2y^2 - 3x)\,dy = 0.$$

$$\left(2xe^y + \frac{2x}{y} + \frac{1}{y^3}\right)dx + \left(x^2e^y - \frac{x^2}{y^2} - \frac{3x}{y^4}\right)dy = 0$$

$$\therefore M_1 = 2xe^y + \frac{2x}{y} + \frac{1}{y^3}, \quad N_1 = x^2e^y - \frac{x^2}{y^2} - \frac{3x}{y^4}.$$

$$\therefore \frac{\partial M_1}{\partial y} = 2xe^y - \frac{2x}{y^2} - \frac{3}{y^4}, \quad \frac{\partial N_1}{\partial x} = 2xe^y - \frac{2x}{y^2} - \frac{3}{y^4}.$$

Here, $\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$, the eq^n is exact.

$\therefore$ The general sol^n is

$$\int M_1\,dx + \int (\text{Terms of } N_1 \text{ not containing } x)\,dy = C.$$

(y-constant)

$$\therefore \int \left(2xe^y + \frac{2x}{y} + \frac{1}{y^3}\right)dx + \int 0\,dy = C$$

$$\therefore x^2e^y + \frac{x^2}{y} + \frac{x}{y^3} = C.$$

Case - III :-

If the D.E. is of the form $f_1(xy)\,y\,dx + f_2(xy)\,x\,dy = 0$ then $I.F. = \frac{1}{Mx - Ny}$, where $M = f_1(xy)\cdot y$, $N = f_2(xy)\,x$, provided $Mx - Ny \neq 0$.

After multiplying with the I.F., the eq^n becomes exact and can be solved using the method of exact differential equations.

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* Examples :-

① Solve $(x^2y^2+2)y\,dx + (2-x^2y^2)x\,dy = 0$.

→

Here, $M = x^2y^3+2y$, $N = 2x-x^3y^2$.

- The eq^n is of the form

$f_1(xy)\,y\,dx + f_2(xy)\,x\,dy = 0$

$$I.F. = \frac{1}{Mx-Ny} = \frac{1}{x^3y^3+2xy-2xy+x^3y^3} = \frac{1}{2x^3y^3}$$

- Multiplying the D.E. by I.F.,

$$\therefore (y^3+2)y\cdot\left(\frac{1}{2x^3y^3}\right)dx + (2-x^2y^2)x\left(\frac{1}{2x^3y^3}\right)dy = 0$$

$$\therefore \frac{1}{2}\left(\frac{1}{x}+\frac{2}{x^3y^2}\right)dx + \frac{1}{2}\left(\frac{2}{x^2y^3}-\frac{1}{y}\right)dy = 0$$

Here, $M_1 = \frac{1}{2}\left(\frac{1}{x}+\frac{2}{x^3y^2}\right)$, $N_1 = \frac{1}{2}\left(\frac{2}{x^2y^3}-\frac{1}{y}\right)$

$$\therefore \frac{\partial M_1}{\partial y} = \frac{1}{2}\left[2\left(-\frac{2}{x^3y^3}\right)\right], \quad \frac{\partial N_1}{\partial x} = \frac{1}{2}\left[2\cdot\left(\frac{-2}{x^3y^3}\right)\right]$$

$$\therefore \frac{\partial M_1}{\partial y} = \frac{-2}{x^3y^3}, \quad \frac{\partial N_1}{\partial x} = \frac{-2}{x^3y^3}.$$

$$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}.$$

∴ The eq^n is exact.

∴ The general solution is

$$\int_{y\ \text{constant}} M_1\,dx + \int (\text{Terms of } N_1 \text{ not containing } x)\,dy = c$$

$$\therefore \int \frac{1}{2}\left(\frac{1}{x}+\frac{2}{x^3y^2}\right)dx + \int \frac{1}{2}\left(\frac{-1}{y}\right)dy = c$$

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$\frac{1}{2}\left(\log x - \frac{1}{x^2y^2}\right) - \frac{1}{2}\log y = c$

$\therefore \log x - \log y - \frac{1}{x^2y^2} = c$

$\therefore \log\left(\frac{x}{y}\right) - \frac{1}{x^2y^2} = c.$

(2) Solve $(xy\sin xy + \cos xy)\, y\, dx + (xy\sin xy - \cos xy)\, x\, dy = 0.$

$\Rightarrow$

Here, $M = xy^2\sin xy + y\cdot\cos xy$, $N = x^2y\sin xy - x\cos xy$.

The eq^n is in the form $f_1(xy)\, y\, dx + f_2(xy)\, x\, dy = 0$.

$$I.F. = \frac{1}{Mx - Ny} = \frac{1}{x^2y^2\sin xy + xy\cos xy - x^2y^2\sin xy + xy\cos xy}$$

$$= \frac{1}{2xy\cos xy}$$

Multiplying the D.E. by the I.F.,

$$\frac{1}{2xy\cos xy}(xy\sin xy + \cos xy)\, y\, dx + \frac{1}{2xy\cos xy}(xy\sin xy - \cos xy)\, x\, dy = 0$$

$$\Rightarrow \left(\frac{y\tan xy}{2} + \frac{1}{2x}\right)dx + \left(\frac{x\tan xy}{2} - \frac{1}{2y}\right)dy = 0$$

Here, $M_1 = \frac{y\tan xy}{2} + \frac{1}{2x}$, $N_1 = \frac{x\tan xy}{2} - \frac{1}{2y}$.

$$\frac{\partial M_1}{\partial y} = \frac{\tan xy + xy\cdot\sec^2 xy}{2}, \quad \frac{\partial N_1}{\partial x} = \frac{\tan xy + xy\sec^2 xy}{2}$$

$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$

$\therefore$ The eq^n is exact.

$\frac{x}{y}\sec xy = e^{2x} = k.$

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Case - IV :-

If the differential equation $M\,dx + N\,dy = 0$ is a homogeneous equation in x & y (degree of each term is same) then

$I.F. = \frac{1}{Mx + Ny}$ provided $Mx + Ny \neq 0$.

- After multiplying with the I.F., the eq^n becomes exact and can be solved using the method of exact differential equations.

* Examples:-

(1) Solve $x^2y\,dx - (x^3 + xy^2)\,dy = 0$

→

Here, $M = x^2y$, $N = -x^3 - xy^2$.

- The differential equation is homogeneous as each term is of degree 3.

$$I.F. = \frac{1}{Mx + Ny} = \frac{1}{x^3y - x^3y + 0 - xy^3} = -\frac{1}{xy^3}$$

- Multiplying the D.E. by the I.F.,

$$\frac{-1}{xy^3}(x^2y)\,dx - \left(\frac{-1}{xy^3}\right)(x^3 + xy^2)\,dy = 0$$

$$\therefore -\frac{x}{y^2}\,dx + \left(\frac{x^2}{y^3} + \frac{1}{y}\right)dy = 0$$

Here, $M_1 = -\frac{x}{y^2}$, $N_1 = \frac{x^2}{y^3} + \frac{1}{y}$.

$$\therefore \frac{\partial M_1}{\partial y} = \frac{2x}{y^3}, \quad \frac{\partial N_1}{\partial x} = \frac{2x}{y^3}$$

$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$, the eq^n is exact.

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- Hence, the general solution is

$\int M_1\,dx + \int$ (Terms of N_1 not containing x) $dy = c$.
y-constant

$\therefore \int \frac{-x}{y^2}\,dx + \int \frac{1}{y}\,dy = c.$

$\therefore -\frac{x^2}{2y^2} + \log y = c.$

(2) Solve $x\frac{dy}{dx} + \frac{y^2}{x} = y.$

$\frac{dy}{dx} + \frac{y^2}{x^2} = \frac{y}{x}$
Not Linear

$\Rightarrow$

$x^2dy + y^2dx = xy\,dx$

$\therefore (y^2 - xy)\,dx + x^2dy = 0$

$\therefore M = y^2 - xy, \quad N = x^2$

The D.E. is homogeneous as degree of each term is 2.

$\therefore I.F. = \frac{1}{Mx + Ny} = \frac{1}{xy^2 - x^2y + x^2y} = \frac{1}{xy^2}.$

- Multiplying the D.E. by I.F.,

$\frac{1}{xy^2}x^2dy + \frac{(y^2 - xy)}{xy^2}dx = 0$

$\therefore \left(\frac{1}{x} - \frac{1}{y}\right)dx + \frac{x}{y^2}dy = 0$

Here, $M_1 = \frac{1}{x} - \frac{1}{y}, \quad N_1 = \frac{x}{y^2}.$

$\frac{\partial M_1}{\partial y} = \frac{1}{y^2}, \quad \frac{\partial N_1}{\partial x} = \frac{1}{y^2}$

$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$, the eq^n is exact.

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- Hence, the general solution is

$\int_{y\text{-constant}} M_1\,dx + \int (\text{Terms of } N_1 \text{ not containing } x)\,dy = c$

$\therefore \int_{y\text{-constant}} \left(\frac{1}{x} - \frac{1}{y}\right) dx + \int 0\,dy = c.$

$\therefore \log x - \frac{x}{y} = c.$

* Case - VI :- Integrating factors by Inspection :-

- Sometimes the integrating factor can be identified by regrouping the terms of the D.E. The following table helps in identifying the I.F. after regrouping the terms.

Sr. No	Group of Terms	Integrating Factor	Exact Differential Equation.
1.	$dx \pm dy$	$\frac{1}{x \pm y}$	$\frac{dx \pm dy}{x \pm y} = d[\log(x \pm y)]$
2.	$y\,dx + x\,dy$	$\frac{1}{2xy}$	$y\,dx + x\,dy = d(xy)$ $2x^2y\,dy + 2xy^2\,dx = d(x^2y^2)$
		$\frac{1}{xy}$	$\frac{y\,dx + x\,dy}{xy} = d[\log(xy)]$
		$\frac{1}{(xy)^n}$	$\frac{y\,dx + x\,dy}{(xy)^n} = d\left[\frac{(xy)^{1-n}}{1-n}\right]$
3.	$y\,dx - x\,dy$	$\frac{1}{y^2}$	$\frac{y\,dx - x\,dy}{y^2} = d\left(\frac{x}{y}\right)$
		$\frac{1}{x^2+y^2}$	$\frac{y\,dx - x\,dy}{x^2+y^2} = d\left[\tan^{-1}\left(\frac{x}{y}\right)\right]$

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Sr. No.	Group of Terms	Integrating Factor	Exact Differential Equation.
		$\frac{1}{x^2}$	$\frac{y\,dx - x\,dy}{x^2} = d\left(\frac{-y}{x}\right)$
		$\frac{1}{xy}$	$\frac{y\,dx - x\,dy}{xy} = d\left[\log\left(\frac{x}{y}\right)\right]$.
4.	$x\,dx \pm y\,dy$	2	$2x\,dx \pm 2y\,dy = d(x^2 \pm y^2)$.
		$\frac{1}{(x^2 \pm y^2)}$	$\frac{2x\,dx \pm 2y\,dy}{x^2 \pm y^2} = d[\log(x^2 \pm y^2)]$
		$\frac{1}{(x^2 \pm y^2)^n}$	$\frac{2x\,dx \pm 2y\,dy}{(x^2 \pm y^2)^n} = d\left[\frac{\log(x^2 \pm y^2)^{1-n}}{2(1-n)}\right]$.
5.	$2y\,dx + x\,dy$	x	$2xy\,dx + x^2\,dy = d(x^2y)$.
6.	$y\,dx + 2x\,dy$	y	$y^2\,dx + 2xy\,dy = d(xy^2)$.
7.	$2y\,dx - x\,dy$	$\frac{x}{y^2}$	$\frac{2xy\,dx - x^2\,dy}{y^2} = d\left(\frac{x^2}{y}\right)$.
8.	$2x\,dy - y\,dx$	$\frac{y}{x^2}$	$\frac{2xy\,dy - y^2\,dx}{x^2} = d\left(\frac{y^2}{x}\right)$.

Examples:-

1) Solve $x\,dy - y\,dx + 2x^3\,dx = 0$.

→ Dividing the eq^n by x^2,

$$\frac{x\,dy - y\,dx}{x^2} + 2x\,dx = 0.$$

$$\therefore d\left(\frac{y}{x}\right) + d(x^2) = 0$$

$$\therefore \frac{y}{x} + x^2 = c.$$

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② Solve $x\,dy - y\,dx = 3x^2(x^2+y^2)\,dx$.

→

Dividing the eq^n by (x^2+y^2),

$$\frac{x\,dy - y\,dx}{x^2+y^2} = 3x^2\,dx$$

$$\Rightarrow d\left[\tan^{-1}\left(\frac{y}{x}\right)\right] = d(x^3).$$

Integrating both the sides,

$$\tan^{-1}\left(\frac{y}{x}\right) = x^3 + C.$$

③ Solve $y(2x^2y + e^x)\,dx - (e^x + y^3)\,dy = 0$.

→

The given D.E. is non exact.

- Rewritting the eq^n, we have

$$2x^2y^2\,dx + e^x y\,dx - e^x\,dy - y^3\,dy$$

$$\therefore 2x^2y^2\,dx - y^3\,dy + e^x(y\,dx - dy) = 0$$

- Dividing by y^2,

$$2x^2\,dx - y\,dy + e^x\left[\frac{y\,dx - dy}{y^2}\right] = 0.$$

$$\frac{2}{3}d(x^3) - \frac{1}{2}d(y^2) + d\left(\frac{e^x}{y}\right) = 0$$

∴ Integrating both the sides, we get the G.S. as

$$\frac{2}{3}x^3 - \frac{1}{2}y^2 + \frac{e^x}{y} = C.$$

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(4) If x^n is an integrating factor of $(y-2x^3)\,dx - x(1-xy)\,dy = 0$ then find n and solve the equation.

→

If x^n is an I.F. then after multiplication with x^n, the eqⁿ becomes exact.

$\therefore (x^n y - 2x^{n+3})\,dx - x^{n+1}(1-xy)\,dy = 0$ is an exact D.E.

where $M = x^n y - 2x^{n+3}$, $N = -x^{n+1}(1-xy)$

$= -x^{n+1} + x^{n+2} y$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

$\therefore x^n = -(n+1)x^n + (n+2)x^{n+1} y$

$\therefore (n+1)x^n + x^n = (n+2)x^{n+1} y$

$\therefore x^n (n+2) - (n+2)x^n \cdot xy = 0$

$\therefore (n+2)x^n [1 - xy] = 0$

$\therefore n+2 = 0$

$\therefore n = -2$.

- Putting $n = -2$ in the eqⁿ,

$(x^{-2} y - 2x)\,dx - x^{-1}(1-xy)\,dy = 0$

$\therefore \left(\frac{y}{x^2} - 2x\right)dx - \left(\frac{1}{x} - y\right)dy = 0$

$\therefore M = \frac{y}{x^2} - 2x$, $N = -\frac{1}{x} + y$

$\frac{\partial M}{\partial y} = \frac{1}{x^2}$, $\frac{\partial N}{\partial x} = \frac{1}{x^2}$.

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the eqⁿ is exact.

Hence, the general solⁿ is

$\int_{y\text{-constant}} M\,dx + \int (\text{Terms of } N \text{ not containing } x)\,dy = c.$

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$$\int \left(\frac{y}{x^2} - 2x\right) dx + \int y\, dy = c$$

$$\therefore -\frac{y}{x} - x^2 + \frac{y^2}{2} = c.$$

⑤ Solve $y\,dx - x\,dy + 3x^2y^2e^{x^3}dx = 0$

→

Rewritting the eq^n as

$$(y + 3x^2y^2e^{x^3})dx - x\,dy = 0$$

Here, $M = y + 3x^2y^2e^{x^3}$, $N = -x$.

$$\frac{\partial M}{\partial y} = 1 + 6x^2ye^{x^3}, \quad \frac{\partial N}{\partial x} = -1$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

∴ The eq^n is not exact.

- Dividing the given eq^n by y^2,

$$\frac{y\,dx - x\,dy}{y^2} + 3x^2e^{x^3}dx = 0$$

$$\therefore d\left(\frac{x}{y}\right) + d(e^{x^3}) = 0.$$

Integrating, we get the general solution,

$$\frac{x}{y} + e^{x^3} = c.$$

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